

TOC LAB EXPERIMENTS(1-3)

Exp 1: Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with a and end with a

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 #include <string.h> 3 4 int main() { 5 char s[100]; 6 printf("Enter string: "); 7 scanf("%s", s); 8 int n = strlen(s); 9 printf(n > 0 && s[0]=='a' && s[n-1]=='a' ? "Accepted" : "Rejected"); 10 return 0; 11 }</pre>		<pre>Enter string: aba Accepted</pre>

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 #include <string.h> 3 4 int main() { 5 char s[100]; 6 printf("Enter string: "); 7 scanf("%s", s); 8 int n = strlen(s); 9 printf(n > 0 && s[0]=='a' && s[n-1]=='a' ? "Accepted" : "Rejected"); 10 return 0; 11 }</pre>		<pre>Enter string: abb Rejected</pre>

Exp 2 : Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with 0 and end with 1

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 #include <string.h> 3 4 int main() { 5 char s[100]; 6 printf("Enter string: "); 7 scanf("%s", s); 8 int n = strlen(s); 9 printf(n > 0 && s[0]=='0' && s[n-1]=='1' ? "Accepted" : "Rejected"); 10 return 0; 11 }</pre>		<pre>Enter string: 001 Accepted</pre>

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 #include <string.h> 3 4 int main() { 5 char s[100]; 6 printf("Enter string: "); 7 scanf("%s", s); 8 int n = strlen(s); 9 printf(n > 0 && s[0]=='0' && s[n-1]=='1' ? "Accepted" : "Rejected"); 10 return 0; 11 }</pre>		<pre>Enter string: 010 Rejected</pre>

Exp 3 : Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0A1$ $A \rightarrow 0A \mid 1A \mid \varepsilon$

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 #include <string.h> 3 4 int main() { 5 char s[100]; 6 printf("Enter string: "); 7 scanf("%s", s); 8 int n = strlen(s); 9 printf(n >= 2 && s[0]=='0' && s[n-1]=='1' ? "Accepted" : "Rejected"); 10 return 0; 11 }</pre>		<pre>Enter string: 01 Accepted</pre>

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 #include <string.h> 3 4 int main() { 5 char s[100]; 6 printf("Enter string: "); 7 scanf("%s", s); 8 int n = strlen(s); 9 printf(n >= 2 && s[0]=='0' && s[n-1]=='1' ? "Accepted" : "Rejected"); 10 return 0; 11 }</pre>		<pre>Enter string: 0 Rejected</pre>